

i. Communicational System $1\text{MHz} = 10\text{kHz}$

ii. Physical World.

Weak Nuclear $- 10^{-13}$ (In Real 10^{-6})

Strong Nuclear $- 1$

Gravitational $- 10^{-39}$ (In Real 10^{-38})

Electromagnetic $- 10^{-3}$ (In Real 10^2)

⇒ Transmitting information ⇒ Propagation.
Sender-Receiver

1. Transmitter

2. Medium / Channel.

3. Receiver.

4. Broadcast.

5. Point to point.

1. Ground Wave

2. Sky Wave.

3. Space Wave.

$$\text{Modulation Index} = \frac{A_m}{A_c} = \frac{E_m}{E_c}$$

$$P \propto \frac{1}{\lambda^2} \quad P \propto V^2$$

P = Power of frequency

$$A_{\text{max}} = A_c + A_m$$

$$A_{\text{min}} = A_c - A_m$$

Height of Radio Signal $h = \sqrt{2} R R$

Height of Transmitting Antenna $h = \frac{d^2}{2R}$

⇒ Area covered : πd^2
 $\pi (\sqrt{2} R h)$

$$2\pi R h$$

"Band"

Transducer: Converts one form of Energy to Another.

Attenuation: Loss of Strength of a Signal while propagating through a Medium.

Amplification: Process of increasing Amp.

Bandwidth: Occupied influence

Modulation: Making Suitable for Transmission

Demodulation: Recover the Original information Signals.

Repeater: Combination of Receiver & Transmitter

$$P = \frac{nE}{t} = \frac{n(\frac{hc}{\lambda})}{t} \propto \frac{V^2}{\lambda^2}$$

$$\frac{\text{Wave Length } (\lambda)}{4} = \frac{\text{Volt (V)}}{2}$$

$$\alpha = \frac{AM}{Ac}$$

Units:

$$1 \text{ MHz} = 10^6 \text{ Hz}$$

$$1 \text{ KHz} = 10^3 \text{ Hz}$$

S_1, S_2 = are Side band frequencies

$$S_{\text{up}} = S_1 = f_c + f_m$$

$$S_{\text{LM}} = S_2 = f_c - f_m$$